

Rational q, t -Catalan Formula

1 Purpose

This item records the conjectural formula and finite checker from Graham Hawkes's paper *A conjectured formula for the rational q, t -Catalan polynomial* (Annals of Combinatorics 28, 749–795, 2024; arXiv:2208.00577).

The source script is

`Conjectures-and-Computations/qt-catalan/qt-conjecture.py`.

The curated script

`code/check_rational_qt_catalan_formula.py`

ports the same finite check into a command-line tool with compact output.

2 Step Coordinates

The checker uses the source script's step-coordinate convention. For a coprime pair (r, n) , put

$$\ell = r - 1.$$

A generated path is a sequence

$$x = (x_0, \dots, x_{\ell-1})$$

constructed from left to right subject to the rational Dyck bound

$$\sum_{a=0}^i x_a \leq \left\lfloor \frac{(i+1)n}{\ell+1} \right\rfloor \quad (0 \leq i < \ell).$$

The script computes a degree statistic incrementally. For

$$\beta(i, j; x) = \sum_{a=i}^j x_a - \frac{n(j-i+1)}{\ell+1},$$

define

$$\gamma(i, j; x) = \begin{cases} \min(x_{i-1}, \lfloor -\beta(i, j; x) \rfloor), & \beta(i, j; x) < 0, \\ \min(x_i, \lfloor \beta(i, j; x) \rfloor), & \beta(i, j; x) > 0, \\ 0, & \beta(i, j; x) = 0. \end{cases}$$

When a new final entry is appended, the degree is increased by the corresponding $\gamma(k, j; x)$ terms with the new endpoint j . This is the source script's statistic used in the monomial comparison.

The maximum area parameter is

$$M = \sum_{i=0}^{\ell-1} \left\lfloor \frac{(i+1)n}{\ell+1} \right\rfloor.$$

For a generated path x , the script uses

$$\text{area}(x) = M - \sum_{p=0}^{\ell-1} \sum_{a=0}^p x_a.$$

3 Conjectural Formula Check

The formula is expressed as a monomial-string identity. The source script first finds the unique closest point to the diagonal used by the conjecture: an index q such that

$$(q+1)n \equiv 1 \pmod{\ell+1},$$

recorded as

$$\left(q, \left\lfloor \frac{(q+1)n}{\ell+1} \right\rfloor \right).$$

The distinguished subfamily T consists of the generated paths satisfying

$$\sum_{a=0}^q x_a = \left\lfloor \frac{(q+1)n}{\ell+1} \right\rfloor.$$

Every generated path contributes one left-side monomial record

$$(\text{area}(x), M - d(x)),$$

where $d(x)$ is the generated degree statistic. Each path in T contributes a monomial string to the right side. If $a = \text{area}(x)$ and $d = d(x)$, then the positive interval is

$$(j, M - d), \quad a \leq j \leq M - a - d,$$

when $a \leq M - a - d$. If $M - a - d < a$, the script records the negative correction interval

$$(j, M - d), \quad M - a - d + 1 \leq j < a.$$

The finite check is the exact multiset identity

$$\text{positive interval multiset} = \text{all path monomials} + \text{negative correction multiset}.$$

4 Reproducible Runs

The source script's built-in example is $(r, n) = (7, 12)$. Running

```
python code/check_rational_qt_catalan_formula.py
```

checks that case. In the curated run it generated 2,652 paths, found closest point $(2, 5)$, and verified

$$2666 \text{ positive terms} = 2652 \text{ all-path terms} + 14 \text{ negative correction terms}.$$

The status was **PASS**.

The script also accepts multiple cases, for example

```
python code/check_rational_qt_catalan_formula.py --case 3/5 --case 5/8 --case 7/12.
```

All listed cases must be coprime. The source README explicitly warns that the conjecture is not expected to hold for non-coprime parameters in general, and the curated checker rejects non-coprime inputs.

5 Status

This item is conjectural. It packages the finite checks for selected relatively prime parameter pairs; it is separate from the computer-assisted proof item for Lemma 2 and Lemma 3 of Section 9.